experiment 7: Text Analytics:

1. Objective

To perform:

- text preprocessing
- natural language processing (NLP)
- TF-IDF analysis

using Python and NLTK library.

---

2. Theory

What is Text Analytics?

Text analytics means extracting meaningful information from text data.

It is used in:

- sentiment analysis
- chatbots
- spam filtering
- search engines
- recommendation systems

---

3. What is NLP?

NLP (Natural Language Processing) is a field of AI that helps computers understand human language.

Examples:

- Google Translate
- ChatGPT
- Voice assistants

---

4. What is NLTK?

NLTK stands for:

Natural Language Toolkit

It is a Python library used for:

- tokenization
- stemming
- POS tagging
- stopword removal
- text processing

Import statement:

import nltk

---

5. Text Preprocessing Steps

1. Tokenization

Breaking text into smaller units called tokens.

Example:

Sentence:

I love AI

Tokens:

['I', 'love', 'AI']

Types:

- Sentence tokenization
- Word tokenization

---

2. Stop Words Removal

Stop words are common words like:

- is
- am
- the
- a
- an

These words do not add important meaning.

---

3. Stemming

Reducing words to root form.

Example:

Original Word	Stem
playing	play
studies	studi

Stemming may not produce meaningful words.

---

4. Lemmatization

Converts word into meaningful base form.

Example:

Original Word	Lemma
studies	study
running	run

Lemmatization is more accurate than stemming.

---

5. POS Tagging

POS means:

Parts Of Speech

Assigns grammatical tags like:

- noun
- verb
- adjective

Example:

Word	POS
boy	noun
runs	verb

---

6. What is TF-IDF?

TF-IDF stands for:

Term Frequency – Inverse Document Frequency

Used to measure importance of a word in a document.

---

7. Term Frequency (TF)

Measures how frequently a word appears in a document.

Formula:

$$TF = \frac{\text{Number of times term appears in document}}{\text{Total number of words in document}}$$

---

8. Inverse Document Frequency (IDF)

Measures importance of word across all documents.

Formula:

$$IDF = \log\left(\frac{\text{Total number of documents}}{\text{Number of documents containing the term}}\right)$$

---

9. TF-IDF Formula

$$TF\text{-}IDF = TF \times IDF$$

Words with high TF-IDF are important words.

---

10. Procedure

Step 1:

Import required libraries

Step 2:

Input sample text

Step 3:

Perform tokenization

Step 4:

Remove stop words

Step 5:

Perform stemming

Step 6:

Perform lemmatization

Step 7:

Perform POS tagging

Step 8:

Calculate TF-IDF

12. Explanation of Code

Tokenization

word_tokenize(text)

Splits sentence into words.

Stopword Removal

stopwords.words('english')

Removes common words.

Stemming

PorterStemmer()

Converts words into root form.

Lemmatization

WordNetLemmatizer()

Converts words into meaningful base form.

POS Tagging

pos_tag(filtered)

Assigns grammatical tags.

TF-IDF

TfidfVectorizer()

Calculates importance of words.

---

13. Difference Between Stemming and Lemmatization

Stemming	**Lemmatization**
Removes suffix	Uses dictionary
Faster	More accurate
May produce invalid words	Produces meaningful words

---

14. Applications of Text Analytics

- Spam detection
- Sentiment analysis
- Chatbots
- Search engines
- Recommendation systems

---

15. Advantages of TF-IDF

- Finds important words
- Removes less useful words
- Improves NLP models

---

16. Conclusion

We successfully performed text preprocessing operations like tokenization, stopword removal, stemming, lemmatization, POS tagging, and implemented TF-IDF for text analysis.

Viva Questions with Answers

1. What is text analytics?

Process of extracting useful information from text.

---

2. What is NLP?

Natural Language Processing.

---

3. What is NLTK?

Python library for natural language processing.

4. What is tokenization?

Breaking text into tokens.

---

5. What are stop words?

Common words with little meaning.

Examples:

- the
- is
- am

---

6. What is stemming?

Reducing word to root form.

---

7. What is lemmatization?

Converting word to meaningful base form.

---

8. Difference between stemming and lemmatization?

Stemming	Lemmatization
Faster	More accurate
May produce invalid words	Produces valid words

---

9. What is POS tagging?

Assigning grammatical tags to words.

---

10. What is TF?

Term Frequency.

---

11. What is IDF?

Inverse Document Frequency.

---

12. What is TF-IDF?

Technique to measure word importance.

---

13. Why remove stop words?

To reduce unnecessary words.

---

14. Which library is used for NLP?

NLTK.

---

15. Which function is used for tokenization?

word_tokenize()

---

16. Which stemmer is used?

PorterStemmer.

---

17. What does TfidfVectorizer do?

Calculates TF-IDF values.

---

18. What is corpus?

Collection of documents.

---

19. What is sentiment analysis?

Finding emotions/opinions from text.

---

20. Applications of text analytics?

- Chatbots
- Spam filtering
- Recommendation systems

Important Viva Tips

If examiner asks:

"Why use TF-IDF?"

Answer:
To identify important words in a document.

"Why remove stop words?"

Answer:
Because they do not provide meaningful information.

"Which is better: stemming or lemmatization?"

Answer:
Lemmatization because it gives meaningful root words.

"Why use POS tagging?"

Answer:
To understand grammatical role of words.

✅ Experiment 4: Data Analytics I – Linear Regression

🔹 **Title**

Implementation of Linear Regression using Python

🔹 **Objective**

- To understand supervised learning
- To implement Linear Regression
- To predict continuous values
- To evaluate regression model performance

🔹 **Theory**

✅ **What is Machine Learning?**

Machine Learning is a branch of AI where systems learn patterns from data and make predictions without explicit programming.

🔹 **Types of Machine Learning**

Type	Description
Supervised Learning	Uses labeled data
Unsupervised Learning	Uses unlabeled data
Reinforcement Learning	Learning through rewards

✅ **What is Supervised Learning?**

In supervised learning:

- Input and output both are known
- Model learns relationship between them

Example:

Study Hours	Marks
2	40
5	70

✅ **What is Regression?**

Regression predicts **continuous numeric values**.

Examples:

- House price prediction
- Salary prediction

- Temperature prediction

✅ **What is Linear Regression?**

Linear Regression finds relationship between:

- Independent variable (X)
- Dependent variable (Y)

It draws a **best fit straight line**.

m

b

-10-8-6-4-2246810-10-5510y-interceptx-intercept

Where:

- y = dependent variable
- x = independent variable
- m = slope
- c = intercept

🔹 **Example**

Study hours increase → marks increase.

This shows a linear relationship.

🔹 **Types of Linear Regression**

Type	**Description**
Simple Linear Regression	One independent variable
Multiple Linear Regression	Multiple independent variables

🔹 **Important Terms**

✅ **Independent Variable**

Input variable used for prediction.

Example:

Study Hours

✅ **Dependent Variable**

Output variable to predict.

Example:

Marks

✅ **Slope (m)**

Shows rate of change.

✅ **Intercept (c)**

Value of y when x = 0.

🔹 **Assumptions of Linear Regression**

1. Linear relationship
2. No multicollinearity
3. Homoscedasticity
4. Normal distribution of residuals

🔹 **Steps in Linear Regression**

1. Import dataset
2. Split dataset
3. Train model
4. Predict values
5. Evaluate model

🔹 **Libraries Used**

Library	Purpose
pandas	Data handling
numpy	Numerical calculations
matplotlib	Graph plotting
sklearn	Machine learning

🔹 **Algorithm / Procedure**

Step 1:

Import libraries

Step 2:

Load dataset

Step 3:

Separate X and Y

Step 4:

Split data into train and test

Step 5:

Train Linear Regression model

Step 6:

Predict test values

Step 7:

Evaluate model performance

🔹 **Explanation of Code**

✅ **train_test_split()**

Splits data into:

- Training data
- Testing data

Example:

test_size=0.2

20% data used for testing.

✅ **LinearRegression()**

Creates Linear Regression model.

✅ **fit()**

Trains the model.

Example:

model.fit(X_train, Y_train)

✅ **predict()**

Predicts output values.

✅ **coef_**

Returns slope value.

✅ **intercept_**

Returns intercept value.

✅ **mean_squared_error()**

Calculates prediction error.

Formula:

$$MSE = \frac{1}{n}\sum(y - \hat{y})^2$$

✅ **RMSE**

Root Mean Squared Error:

$$RMSE = \sqrt{MSE}$$

Lower RMSE = better model.

✅ **R² Score**

Measures accuracy of model.

Range:

0 to 1

Closer to 1 = better model.

Formula:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

🔹 **Graph Interpretation**

Scatter Plot

Shows actual data points.

Regression Line

Shows predicted relationship.

🔹 **Expected Output**

- Model trained successfully
- Predicted salary values generated
- Regression line plotted
- RMSE and R^2 score calculated

🔹 **Applications**

1. House price prediction
2. Sales forecasting
3. Weather prediction
4. Stock analysis
5. Salary prediction

🔹 **Advantages**

- Simple and easy

- Fast training
- Easy interpretation

◆ **Disadvantages**

- Works only for linear relationships
- Sensitive to outliers
- Cannot handle complex patterns well

◆ **Conclusion**

In this experiment, Linear Regression was implemented to predict continuous values. The model was trained, tested, and evaluated using RMSE and R^2 score.

◆ **Frequently Asked Viva Questions**

✅ **1. What is Machine Learning?**

Machine learning enables systems to learn from data.

✅ **2. What is supervised learning?**

Learning using labeled data.

✅ **3. What is regression?**

Technique used to predict continuous values.

✅ **4. What is Linear Regression?**

$$y = mx + c$$

$$m$$

$$b$$

-10-8-6-4-2246810-10-5510y-interceptx-intercept

Algorithm that models linear relationship.

✅ **5. Difference between regression and classification?**

Regression	Classification
Predicts continuous values	Predicts categories

✅ **6. What is dependent variable?**

Output variable to predict.

✅ **7. What is independent variable?**

Input variable.

✅ **8. What is slope?**

Rate of change of line.

✅ **9. What is intercept?**

Value when x = 0.

✅ **10. What is overfitting?**

Model learns training data too much.

✅ **11. What is underfitting?**

Model fails to learn patterns.

✅ **12. What is RMSE?**

$$RMSE = \sqrt{MSE}$$

Measure of prediction error.

✅ **13. What is R^2 score?**

Measures goodness of fit.

✅ **14. What does fit() do?**

Trains the model.

✅ **15. What does predict() do?**

Predicts output values.

✅ **16. Why split train and test data?**

To evaluate model performance on unseen data.

✅ **17. What is the use of matplotlib?**

Used for data visualization.

✅ Experiment 5: Data Analytics II – Logistic Regression

🔹 **Title**

Implementation of Logistic Regression using Python

🔹 **Objective**

- To understand classification algorithms
- To implement Logistic Regression
- To classify categorical outputs
- To evaluate classification models

🔹 **Theory**

✅ **What is Classification?**

Classification is a supervised learning technique used to predict categorical outputs.

Examples:

- Spam / Not Spam
- Pass / Fail
- Disease / No Disease

🔹 **Difference Between Regression and Classification**

Regression	Classification
Predicts continuous values	Predicts categories
Example: Salary	Example: Pass/Fail

✅ **What is Logistic Regression?**

Logistic Regression is a supervised machine learning algorithm used for:

- Binary classification
- Probability prediction

It predicts outputs like:

0 or 1
Yes or No
True or False

🔹 **Why "Regression" Name?**

Even though used for classification, it uses a regression equation internally.

🔹 **Sigmoid Function**

Logistic Regression uses the **Sigmoid Function**.

Formula:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

🔹 Output Range of Sigmoid Function

Range:

0 to 1

Probability close to:

- 0 → Class 0
- 1 → Class 1

🔹 Logistic Regression Equation

$$z = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_n x_n$$

Then sigmoid applied:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

🔹 Applications of Logistic Regression

1. Email spam detection
2. Disease prediction
3. Fraud detection
4. Customer churn prediction
5. Pass/fail prediction

🔹 Types of Logistic Regression

Type	**Description**
Binary Logistic Regression	Two classes
Multinomial Logistic Regression	Multiple classes
Ordinal Logistic Regression	Ordered categories

🔹 Important Terms

✅ Binary Classification

Only two classes:

0 and 1

✅ Probability

Chance of occurrence between 0 and 1.

✅ Threshold

Usually:

0.5

If probability > 0.5 → Class 1

Else → Class 0

🔹 **Confusion Matrix**

A table used to evaluate classification performance.

✅ **Structure of Confusion Matrix**

Actual / Predicted	**Positive**	**Negative**
Positive	TP	FN
Negative	FP	TN

🔹 **Terms Explained**

✅ **True Positive (TP)**

Correctly predicted positive.

✅ **True Negative (TN)**

Correctly predicted negative.

✅ **False Positive (FP)**

Incorrectly predicted positive.

✅ **False Negative (FN)**

Incorrectly predicted negative.

🔹 **Evaluation Metrics**

✅ **1. Accuracy**

Measures overall correctness.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

✅ **2. Precision**

Measures correct positive predictions.

Formula:

$$Precision = \frac{TP}{TP + FP}$$

✅ **3. Recall**

Measures ability to find positives.

Formula:

$$Recall = \frac{TP}{TP + FN}$$

✅ **4. F1-Score**

Harmonic mean of precision and recall.

Formula:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

🔹 **Libraries Used**

Library	Purpose
pandas	Data handling
numpy	Numerical operations
sklearn	Machine learning
matplotlib	Visualization

🔹 **Algorithm / Procedure**

Step 1:

Import libraries

Step 2:

Load dataset

Step 3:

Separate input and output

Step 4:

Split dataset

Step 5:

Train Logistic Regression model

Step 6:

Predict outputs

Step 7:

Generate confusion matrix

Step 8:

Evaluate accuracy and metrics

Explanation of Code

✅ **LogisticRegression()**

Creates Logistic Regression model.

✅ **fit()**

Trains the model using training data.

✅ **predict()**

Predicts class labels.

✅ **confusion_matrix()**

Creates confusion matrix.

✅ **accuracy_score()**

Calculates model accuracy.

✅ **classification_report()**

Displays:

- Precision
- Recall
- F1-score
- Support

🔹 **Expected Output**

- Logistic Regression model trained
- Predictions generated
- Confusion matrix displayed
- Accuracy and metrics calculated

🔹 **Applications**

1. Spam email detection
2. Fraud detection
3. Disease diagnosis
4. Credit card approval

5. Customer behavior prediction

◆ **Advantages**

- Simple and efficient
- Fast training
- Works well for binary classification
- Easy interpretation

◆ **Disadvantages**

- Cannot model complex relationships well
- Sensitive to outliers
- Requires large datasets for better accuracy

◆ **Conclusion**

In this experiment, Logistic Regression was implemented for binary classification. The model performance was evaluated using confusion matrix, accuracy, precision, recall, and F1-score.

◆ **Frequently Asked Viva Questions**

✅ **1. What is classification?**

Predicting categorical outputs.

✅ **2. What is Logistic Regression?**

Classification algorithm used for binary classification.

✅ **3. Difference between Linear and Logistic Regression?**

Linear Regression	Logistic Regression
Predicts continuous values	Predicts categories

✅ **4. What is Sigmoid Function?**

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Converts output into probability.

✅ **5. Output range of sigmoid function?**

0 to 1.

✅ **6. What is confusion matrix?**

Table used to evaluate classification model.

✅ **7. What is TP?**

True Positive.

✅ **8. What is FP?**

False Positive.

✅ **9. What is FN?**

False Negative.

✅ **10. What is TN?**

True Negative.

✅ **11. What is accuracy?**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

✅ **12. What is precision?**

$$Precision = \frac{TP}{TP + FP}$$

✅ **13. What is recall?**

$$Recall = \frac{TP}{TP + FN}$$

✅ **14. What is F1-score?**

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

✅ **15. Why use train-test split?**

To evaluate model on unseen data.

✅ **16. What is binary classification?**

Classification with two classes.

✅ **17. What is overfitting?**

Model memorizes training data and performs poorly on new data.

✅ **18. What is underfitting?**

Model fails to learn data patterns.

✅ **19. Which library is used for ML in Python?**

scikit-learn

✅ **20. What is the use of classification_report()?**

Displays precision, recall, F1-score, and support.

Experiment 6: Data Analytics III – Naïve Bayes Classification

Aim

To implement the **Naïve Bayes Classification Algorithm** using Python on the Iris dataset and calculate:

- Confusion Matrix
- TP, TN, FP, FN
- Accuracy
- Error Rate
- Precision
- Recall

---

1. What is Naïve Bayes?

Naïve Bayes is a **supervised machine learning classification algorithm** based on **Bayes Theorem**.

It predicts the class of data using probability.

It is called **"Naïve"** because it assumes that all features are **independent** of each other.

Example:

- Sepal length does not depend on petal width.
- Petal length does not depend on sepal width.

This assumption is not always true, but the algorithm still works very well.

---

2. Bayes Theorem

Bayes theorem formula:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

$$P(A)$$

$$P(B \mid A)$$

$$P(B \mid \neg A)$$

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} \approx 0.68,\ P(B) \approx 0.25$$

P(B)=0.25P(B|A)P(A)=0.17P(A|B)~0.68Posterior = useful evidence / total evidence

Where:

- $P(A \mid B)$= Probability of A given B
- $P(B \mid A)$= Probability of B given A
- $P(A)$= Prior probability
- $P(B)$= Evidence probability

In classification:

- A = class label
- B = feature values

The class with highest probability is selected.

---

3. Types of Naïve Bayes

1. Gaussian Naïve Bayes

Used for **continuous numerical data**.

Example:

- Iris dataset
- Height, weight, age

2. Multinomial Naïve Bayes

Used for:

- Text classification
- Word frequency

3. Bernoulli Naïve Bayes

Used for:

- Binary features (0/1)

---

4. Why Gaussian Naïve Bayes for Iris?

Iris dataset contains:

- Sepal length
- Sepal width
- Petal length
- Petal width

These are continuous numeric values.

Hence we use:

Gaussian Naïve Bayes

---

5. Iris Dataset

The Iris dataset has:

- 150 rows
- 4 features
- 3 flower classes

Classes:

1. Iris-setosa
2. Iris-versicolor
3. Iris-virginica

---

6. Steps of Experiment

Step 1:

Import libraries

Step 2:

Load Iris dataset

Step 3:

Split dataset into:

- Training data
- Testing data

Step 4:

Train Gaussian Naïve Bayes model

Step 5:

Predict output

Step 6:

Create confusion matrix

Step 7:

Calculate:

- Accuracy
- Precision
- Recall

8. Explanation of Code

Import Libraries

import pandas as pd

Used for data handling.

from sklearn.naive_bayes import GaussianNB

Imports Gaussian Naïve Bayes model.

Load Dataset

iris = load_iris()

Loads inbuilt Iris dataset.

Features and Target

X = iris.data
y = iris.target

- X = independent variables
- y = target/output class

Train Test Split

train_test_split()

Splits dataset:

- 70% training
- 30% testing

Create Model

model = GaussianNB()

Creates Naïve Bayes classifier.

Train Model

model.fit(X_train, y_train)

Trains model using training data.

Prediction

y_pred = model.predict(X_test)

Predicts output for testing data.

---

9. Confusion Matrix

Confusion matrix compares:

- Actual values
- Predicted values

Example:

Actual/Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

---

10. Important Terms

True Positive (TP)

Model predicts positive correctly.

True Negative (TN)

Model predicts negative correctly.

False Positive (FP)

Model predicts positive incorrectly.

False Negative (FN)

Model predicts negative incorrectly.

---

11. Accuracy Formula

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy means:
How many predictions are correct.

---

12. Error Rate

$$Error\ Rate = \frac{FP + FN}{TP + TN + FP + FN}$$

---

13. Precision

$$Precision = \frac{TP}{TP + FP}$$

Precision tells:
Out of predicted positives, how many are actually positive.

---

14. Recall

$$Recall = \frac{TP}{TP + FN}$$

Recall tells:
Out of actual positives, how many are predicted correctly.

---

15. Advantages of Naïve Bayes

- Simple and fast
- Works well on large datasets
- Good for text classification
- Requires less training data

---

16. Disadvantages

- Assumes feature independence
- Less accurate when features are dependent

17. Applications

- Spam filtering
- Sentiment analysis
- Email classification
- Medical diagnosis
- Recommendation systems

---

18. Conclusion

We implemented the Gaussian Naïve Bayes algorithm on the Iris dataset and evaluated model performance using confusion matrix, accuracy, precision, and recall.

Viva Questions with Answers

1. What is Naïve Bayes?

Naïve Bayes is a supervised machine learning classification algorithm based on Bayes theorem.

---

2. Why is it called "Naïve"?

Because it assumes all features are independent.

---

3. Which theorem is used in Naïve Bayes?

Bayes theorem.

---

4. What are the types of Naïve Bayes?

1. Gaussian
2. Multinomial
3. Bernoulli

---

5. Which Naïve Bayes is used for Iris dataset?

Gaussian Naïve Bayes.

---

6. Why Gaussian Naïve Bayes?

Because Iris dataset contains continuous numerical values.

---

7. What is supervised learning?

Learning using input and output labeled data.

---

8. What is classification?

Predicting categorical output labels.

---

9. What is confusion matrix?

A table used to evaluate classification performance.

---

10. Define TP.

Correctly predicted positive values.

---

11. Define TN.

Correctly predicted negative values.

---

12. Define FP.

Incorrectly predicted positive values.

---

13. Define FN.

Incorrectly predicted negative values.

---

14. What is accuracy?

Ratio of correctly predicted observations to total observations.

---

15. What is precision?

Correct positive predictions divided by total predicted positives.

---

16. What is recall?

Correct positive predictions divided by total actual positives.

---

17. What is Iris dataset?

A flower dataset containing 3 flower classes and 4 features.

---

18. What are features in Iris dataset?

- Sepal length
- Sepal width
- Petal length
- Petal width

---

19. What is model training?

Teaching model using training dataset.

---

20. What is testing data?

Data used to evaluate model performance.

---

21. Which library is used for Naïve Bayes?

Scikit-learn.

---

22. What is Gaussian distribution?

Normal distribution with bell-shaped curve.

---

23. What does fit() do?

Trains the model.

---

24. What does predict() do?

Predicts output labels.

---

25. What is machine learning?

A field where machines learn patterns from data.

Important Viva Tip

Examiner may ask:

"Why Naïve Bayes works fast?"

Answer:
Because it assumes feature independence and performs probability calculations efficiently.

"Why use Iris dataset?"

Answer:
It is simple, clean, and commonly used for classification experiments.

"Difference between Logistic Regression and Naïve Bayes?"

Answer:

- Logistic Regression predicts using decision boundary.
- Naïve Bayes predicts using probabilities and Bayes theorem.